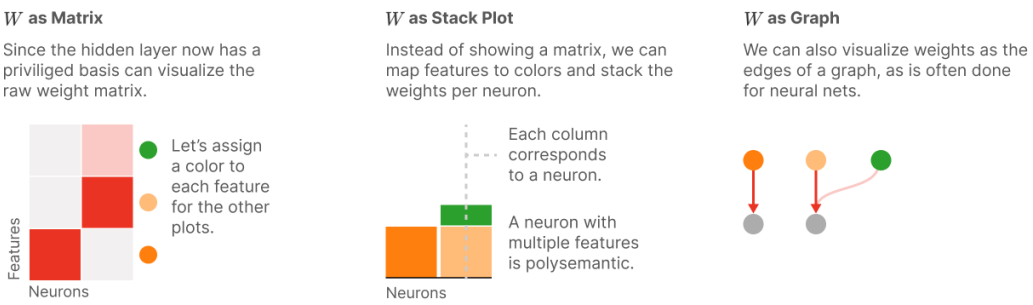
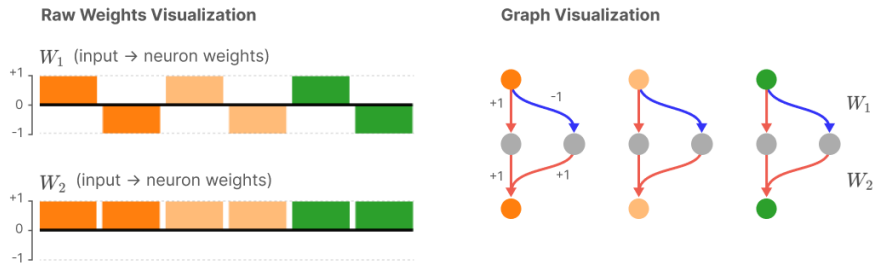


Let's look at what happens when we train a model with $n = 3$ features to perform absolute value on $m = 6$ hidden layer neurons. Without superposition, the model needs two hidden layer neurons to implement absolute value on one feature.

The resulting model – modulo a subtle issue about rescaling input and output weights[15] – performs absolute value exactly as one might expect. For each input feature x_i, it constructs a "positive side" neuron $\mathrm{ReLU}(x_i)$ and a "negative side" neuron $\mathrm{ReLU}(-x_i)$. It then adds these together to compute absolute value:



Superposition vs Sparsity

We've seen that – as expected – our toy model can learn to implement absolute value. But can it use superposition to compute absolute value for more features? To test this, we train models with $n = 100$ features and $m = 40$ neurons and a feature importance curve $I_i = 0.8^i$, varying feature sparsity.[16]

A couple of notes on visualization: Since we're primarily interested in understanding superposition and polysemantic neurons, we'll show a stacked weight plot of the absolute values of weights. The features are colored by superposition. To make the diagrams easier to read, neurons are faintly colored based on how polysemantic they are (as judged by eye based on the plots). Neuron order is sorted by the importance of the largest feature.

ReLU Hidden Layer Toy Model on Absolute Value Task

$n = 100;\ m = 40;\ I_i = 0.8^i$

$1 - S = 1.0$

In the dense regime, all neurons are monosemantic, dedicated to a single feature.

$1 - S = 0.3$

Neruons continue to be monosemantic to moderate sparsity levels.

$1 - S = 0.1$

Eventually, we start to see a few slightly polysemantic neurons.

$1 - S = 0.03$

As sparsity increases further, we see a small number of highly polysemantic neurons representing low importance features.

$1 - S = 0.01$

The number of polysemantic neurons grows...

$1 - S = 0.003$

And they become even more polysemantic...

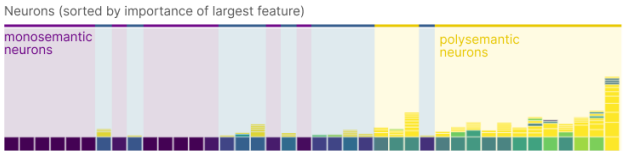
$1 - S = 0.001$

Eventually, all neurons are highly polysemantic.

Much like we saw in the ReLU hidden layer models, these results demonstrate that activation functions, under the right circumstances, create a privileged basis and cause features to align with basis dimensions. In the dense regime, we end up with each neuron representing a single feature, and we can read feature values directly off of neuron activations.

However, once the features become sufficiently sparse, this model, too, uses superposition to represent more features than it has neurons. This result is notable because it demonstrates the ability of neural networks to **perform computation** even on data that is represented in superposition.[17] Remember that the model is required to use the hidden layer ReLU in order to compute an absolute value; gradient descent manages to find solutions that usefully approximate the computation even when each neuron encodes a mix of multiple features.

Focusing on the intermediate sparsity regimes, we find several additional qualitative behaviors that we find fascinatingly reminiscent of behavior that has been observed in real, full-scale neural networks:



$1 - S = 0.01$

Neurons can be monosemantic and polysemantic in the same model.

To begin, we find that in some regimes, **many** of the model's neurons will encode pure features, but a subset of them will be highly polysemantic. This is similar to the phase change we saw earlier in the Relu output model. However, in that case, the phase change was with respect to features, with more important features not being put in superposition. In this experiment, the neurons don't have any intrinsic importance, but we see that the neurons representing the most important features (on the left) tend to be monosemantic.

We find this to bear a suggestive resemblance to some previous work in vision models, which found some layers that contained "mostly pure" feature neurons, but with some neurons representing additional features on a different scale.

We also note that many neurons appear to be associated with a single "primary" feature – encoded by a relatively large weight – coupled with one or more "secondary" features encoded with smaller-magnitude weights to that neuron. If we were to observe the activations of such a neuron over a range of input examples, we would find that the largest activations of that neuron were all or nearly-all associated with the presence of the "primary" feature, but that the lower-magnitude activations were much more polysemantic.

Intriguingly, that description closely matches what researchers have found in previous work on language models [2] – many neurons appear interpretable when we examine their strongest activations over a dataset, but can be shown on further investigation to activate for other meanings or patterns, often at a lower magnitude. While only suggestive, the ability of our toy model to reproduce these qualitative features of larger neural networks offers an exciting hint that these models are illuminating general phenomena.

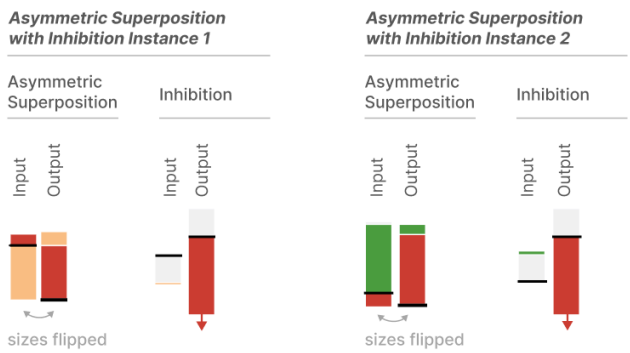
The Asymmetric Superposition Motif

If neural networks can perform computation in superposition, a natural question is to ask how exactly they're doing so. What does that look like mechanically, in terms of the weights? In this subsection, we'll (mostly) work through one such model and see an interesting motif of ***asymmetric superposition***. (We use the term "motif" in the sense of the original circuit thread, inspired by its use in systems biology [36].)

The model we're trying to understand is shown below on the left, visualized as a neuron weight stack plot, with features corresponding to colors. The model is only doing a limited amount of superposition, and many of the weights can be understood as simply implementing absolute value in the expected way.

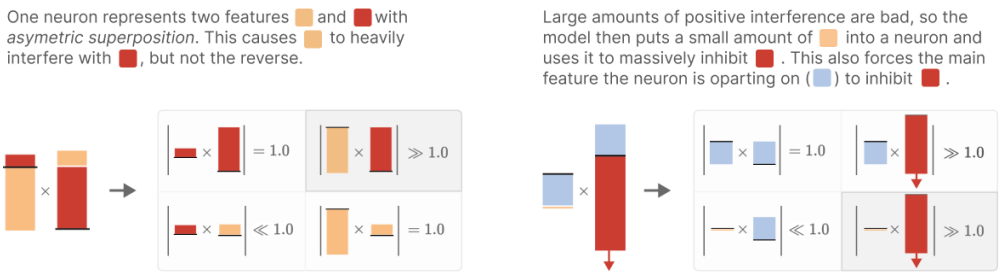
However, there are a few neurons doing something else...

These other neurons implement two instances of asymmetric superposition and inhibition. Each instance consists of two neurons:



One neuron does *asymmetric superposition*. In normal superposition, one might store features with equal weights (eg. $W = [1, -1]$) and then have equal output weights ($W = [1, 1]$). In asymmetric superposition, one stores the features with different magnitudes (eg. $W = [2, -\frac{1}{2}]$) and then has reciprocal output weights (eg. $W = [\frac{1}{2}, 2]$). This causes one feature to heavily interfere with the other, but avoid the other interfering with the first!

To avoid the consequences of that interference, the model has another neuron heavily inhibit the feature in the case where there would have been positive interference. This essentially converts positive interference (which could greatly increase the loss) into negative interference (which has limited consequences due to the output ReLU).



There are a few other weights this doesn't explain. (We believe they're effectively small conditional biases.) But this asymmetric superposition and inhibition pattern appears to be the primary story.

The Strategic Picture of Superposition

Although superposition is scientifically interesting, much of our interest comes from a pragmatic motivation: we believe that superposition is deeply connected to the challenge of using interpretability to make claims about the safety of AI systems. In particular, it is a clear challenge to the most promising path we see to be able to say that neural networks won't perform certain harmful behaviors or to catch "unknown unknowns" safety problems. This is because superposition is deeply linked to the ability to identify and enumerate over all features in a model, and the ability to enumerate over all features would be a powerful primitive for making claims about model behavior.

We begin this section by describing how "solving superposition" in a certain sense is equivalent to many strong interpretability properties which might be useful for safety. Next, we'll describe three high level strategies one might take to "solving superposition." Finally, we'll describe a few other additional strategic considerations.

Safety, Interpretability, & "Solving Superposition"

We'd like a way to have confidence that models will never do certain behaviors such as "deliberately deceive" or "manipulate." Today, it's unclear how one might show this, but we believe a promising tool would be the ability to *identify and enumerate over all features*. The ability to have a universal quantifier over the fundamental units of neural network computation is a significant step towards saying that certain types of circuits don't exist.[18] It also seems like a powerful tool for addressing "unknown unknowns", since it's a way that one can fully cover network behavior, in a sense.

How does this relate to superposition? It turns out that the ability to enumerate over features is deeply intertwined with superposition. One way to see this is to imagine a neural network with a privileged basis and without superposition (like the monosemantic neurons found in early InceptionV1, e.g. [1]): features would simply correspond to neurons, and you could enumerate over features by enumerating over neurons.[19] The connection also goes the other way: if one has the ability to enumerate over features, one can perform compressed sensing using the feature directions to (with high probability) "unfold" a superposition models activations into those of a larger, non-superposition model.

For this reason, we'll call any method that gives us the ability to enumerate over features – and equivalently, unfold activations – a "solution to superposition". Any solution is on the table, from creating models that just don't have superposition, to identifying what directions correspond to features after the fact. We'll discuss the space of possibilities shortly.

We've motivated "solving superposition" in terms of feature enumeration, but it's worth noting that it's equivalent to (or necessary for) many other interpretability properties one might care about: